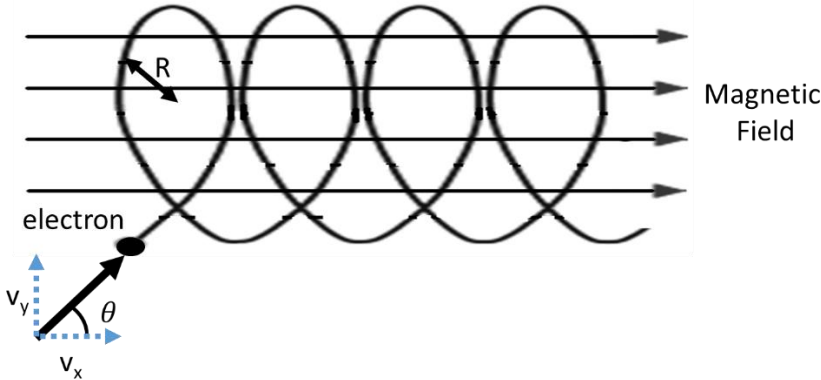


5ai	The field cannot be magnetic since a magnetic force <u>only acts on a charged particle moving at an angle</u> (not parallel or antiparallel) to the magnetic field.	B1
5aii	The field cannot be electric because a stationary electron in an electric field will <u>experience a force in a direction opposite to the field</u> and not in the direction of the field.	B1
5aiii	The field has to be gravitational because a mass (electron in this case) will <u>experience a force in the direction of the gravitational field</u> .	B1
5bi	To maintain equilibrium, the electric force and the gravitational force acting on the electron must be oppositely directed, and	

	$Eq = mg$ $E = \frac{(9.11 \times 10^{-31})(9.81)}{1.60 \times 10^{-19}}$ $E = 5.6 \times 10^{-11} \text{ N C}^{-1}$	C1 A1
5bii	The calculation in (b)(i) shows that only a <u>very small electric field is required to overcome the effects of a gravitational field</u> . In other words, the electric force is typically very much bigger than the gravitational force.	B1
5ci	 The electron's velocity v can be resolved into the vertical and horizontal components v_y and v_x respectively. The vertical component v_y causes the electron to <u>experience a magnetic force perpendicular to the magnetic field and the velocity</u>, causing it to describe a circular path in the magnetic field. The horizontal component v_x is parallel to the magnetic field, therefore this component <u>does not experience any magnetic force</u>. This leads to the electron moving along the field lines. These two effects together result in the electron describing a helical path in the magnetic field.	B1 B1
5cii	$v_y = v \sin \theta = (7.0 \times 10^6) \left(\frac{3}{5} \right)$ $= 4.2 \times 10^6 \text{ m s}^{-1}$	M1 A0
5ciii	For circular motion of electron, $Bqv_y = \frac{mv_y^2}{R}$ $R = \frac{mv_y}{Be} = \frac{(9.11 \times 10^{-31})(4.2 \times 10^6)}{(3.14 \times 10^{-5})(1.6 \times 10^{-19})}$ $R = 0.762 \text{ m}$	C1 A1